第二章 线性回归

Linear Regression model

$$y = 33.73 + 0.516x$$

Linear Regression model

- If a regression model has only two unknown parameters, then it is a binary regression model
- If there are more than two parameters, then it is a multiple regression model

BINARY REGRESSION MODEL

A binary regression model takes the form:

$$y_i = \beta_o + \beta_1 x_{1i} + \varepsilon_i$$

where y = dependent variable

x = one explanatory variable

β_o and β_1 = parameters to be estimated

ε = random error or disturbance

In order to estimate parameters, specific assumptions regarding the probability distribution of ε must be made. These assumptions are very basic to any statistical regression analysis:

Assumptions of Linear Regression models

- **Assumptions 1:**

  Continuous Dependent Variable Y

  Y -- Count variables: Poisson and negative binomial regression

  Y -- Nominal scale variables: Discrete outcome models

  Y -- Ordered scale variables: Ordered regression models

- **Assumptions 2:**

  Linear-in-Parameters Relationship between Y and X

  $$y_i = \beta_0 + \beta_1 x_{1i} + \varepsilon_i$$

  $$\mathbf{Y}_{n \times 1} = \mathbf{X}_{n \times p}\boldsymbol{\beta}_{p \times 1} + \boldsymbol{\varepsilon}_{n \times 1}$$

Assumptions of Linear Regression models

- **Assumptions 3:**

  Observations Independently and Randomly Sampled

  Independence requires that the probability that an observation is selected is unaffected by other observations selected into the sample.

  Other sampling schemes such as stratified and cluster samples can be accommodated in the regression modeling framework with **corrective measures**

Assumptions of Linear Regression models

- **Assumptions 4:**

Uncertain Relationship between Variables

$$Y_i = \beta_0 + \beta_1 X_{1i} \boxed{+ \varepsilon_i}$$

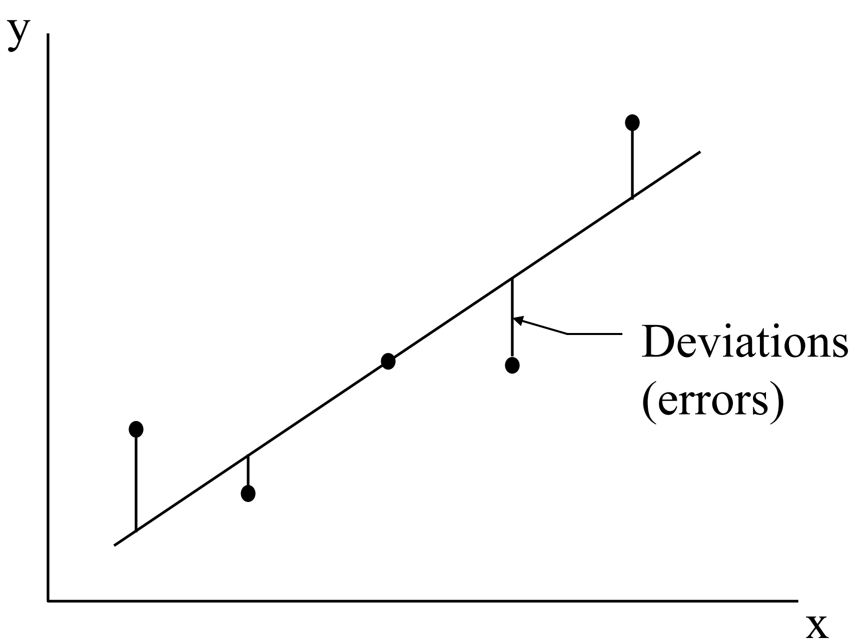
Variables that were **omitted** from the model

Measurement errors in the dependent variable, or the imprecision in measuring Y.

Random variation inherent in the underlying data-generating process.

Least Squares Estimation

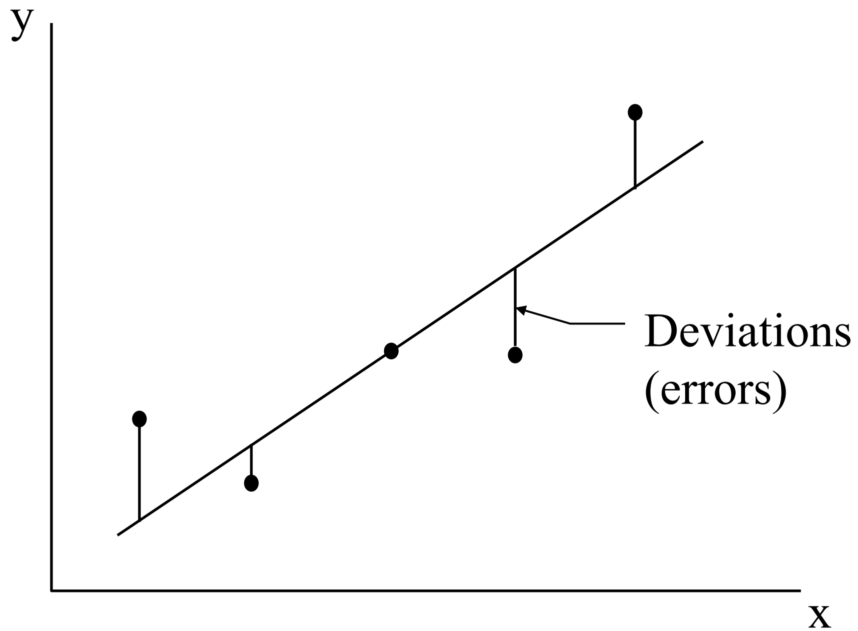
In the figure, the vertical line segments represent deviations of the observation points from the line. One can find many lines for the sum of deviations (errors) is equal to zero. But it can be shown that there is one and only one line for which the SUM of SQUARED DEVIATIONS is a minimum. This sum is called the sum of squared errors (SSE).



$$y = \beta_o + \beta_1 x + \varepsilon$$

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$$

Least Squares Estimation



$$y = \beta_o + \beta_1 x + \varepsilon$$

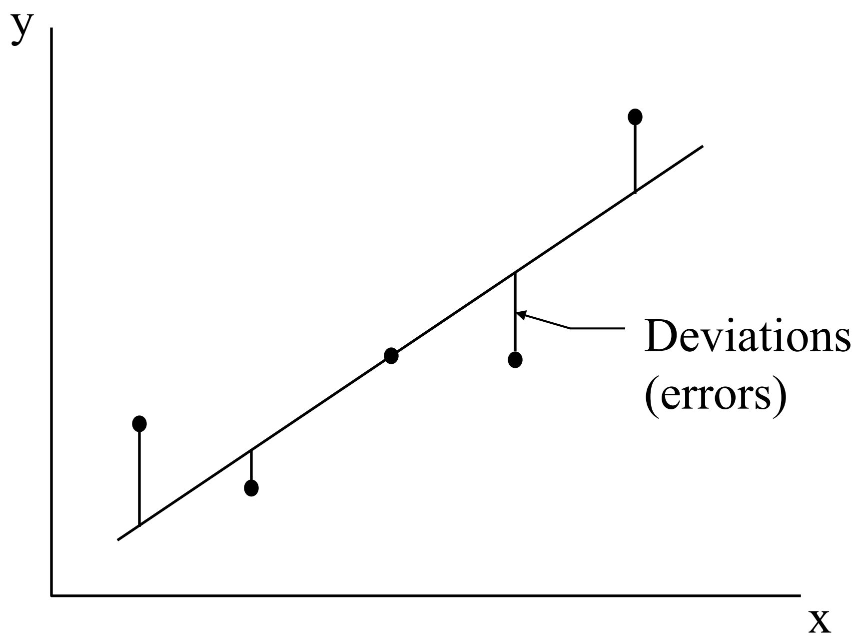
$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$$

Thus, $\hat{y}$ is an estimator of the mean value of y, E(y), and a predictor of some future value of y. $\hat{\beta}_o$ and $\hat{\beta}_1$ are least squares estimators of β_o and β_1 respectively.

For a given data point, (x_i, y_i), the observed value of y is y_i and the predicted value of y would be obtained by substituting x_i into the prediction equation:

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$$

Least Squares Estimation



$$y = \beta_o + \beta_1 x + \varepsilon$$

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$$

The deviation of the ith value of y from its predicted value is:

$$(y_i - \hat{y}_i) = [y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_i)]$$

Then, the sum of squares of the deviations of the y values about their predicted values for all of the observations (n data points):

$$SSE = \sum_{i=1}^{n} [y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_i)]^2$$

Least Squares Estimation

OLS seeks a solution that minimizes the function *Q:*

$$Q_{\min} = \sum_{i=1}^{n}\left(Y_i - \hat{Y}_i\right)^2_{\min} = \sum_{i=1}^{n}\left(Y_i - (\beta_0 + \beta_1 X_i)\right)^2_{\min} = \sum_{i=1}^{n}\left(Y_i - \beta_0 - \beta_1 X_i\right)^2_{\min}$$

By setting the partial derivatives of Q with respect to β_o and β_1 equal to zero, the least squares estimated parameters β_o and β_1 are obtained:

$$\frac{\partial Q}{\partial \beta_0} = -2\sum_{i=1}^{n}\left(Y_i - \beta_0 - \beta_1 X_i\right) = 0$$

$$\frac{\partial Q}{\partial \beta_1} = -2\sum_{i=1}^{n} X_i\left(Y_i - \beta_0 - \beta_1 X_i\right) = 0$$

Least Squares Estimation

Solving these equations using B_0 and B_1 to denote the estimates of β_o and β_1 , respectively, and rearranging terms yields:

$$\sum_{i=1}^{n} Y_i = nB_0 + B_1 \sum_{i=1}^{n} X_i \, , \quad \sum_{i=1}^{n} X_i Y_i = B_0 \sum_{i=1}^{n} X_i + B_1 \sum_{i=1}^{n} X_i^2$$

Solving simultaneously for the betas yields:

$$B_1 = \frac{\sum_{i=1}^{n} \left(X_i - \overline{X}\right)\left(Y_i - \overline{Y}\right)}{\sum_{i=1}^{n} \left(X_i - \overline{X}\right)^2} \, , \quad B_0 = \frac{1}{n}\left(\sum_{i=1}^{n} Y_i - B_1 \sum_{i=1}^{n} X_i \right) = \overline{Y} - B_1 \overline{X}$$

Least Squares Estimation

A database consisting of 121 observations is available to study annual average daily traffic (*AADT*) in Minnesota. Variables available in this database, and their abbreviations, are provided in Table 3.2. A simple regression model is estimated using least squares estimated parameters as a starting point. The starter specification is based on a model that has *AADT* as a function of *CNTYPOP*, *NUMLANES*, and *FUNCTIONALCLASS*: $AADT = \beta_0 + \beta_1(CNTYPOP) + \beta_2(NUMLANES) + \beta_3(FUNCTIONALCLASS)$ + disturbance.

Variables Collected on Minnesota Roadways

Variable No.	*Abbreviation*: Variable Description
1	*AADT*: Average annual daily traffic in vehicles per day
2	*CNTYPOP*: Population of county in which road section is located (proxy for nearby population density)
3	*NUMLANES*: Number of lanes in road section
4	*WIDTHLANES*: Width of road section in feet
5	*ACCESSCONTROL*: 1 for access controlled facility; 2 for no access control
6	*FUNCTIONALCLASS*: Road sectional functional classifications; 1 = rural interstate, 2 = rural non-interstate, 3 = urban interstate, 4 = urban non-interstate
7	*TRUCKROUTE:* Truck restriction conditions: 1 = no truck restrictions, 2 = tonnage restrictions, 3 = time of day restrictions, 4 = tonnage and time of day restrictions, 5 = no trucks
8	*LOCALE*: Land-use designation: 1 = rural, 2 = urban with population ≤ 50,000, 3 = urban with population > 50,000

Least Squares Estimation

Least Squares Estimated Parameters (Example 3.1)

Parameter	Parameter Estimate
Intercept	–26234.490
CNTYPOP	0.029
NUMLANES	9953.676
FUNCLASS1	885.384
FUNCLASS2	4127.560
FUNCLASS3	35453.679

- For each additional 1000 people in the local county population, there is an estimated 29 additional AADT.
- For each lane there is an estimated 9954 AADT.
- Urban interstates are associated with an estimated 35,454 AADT more than non-interstates

Maximum Likelihood Estimation

- Another popular and sometimes useful statistical estimation method is called maximum likelihood estimation, which results in the maximum likelihood estimates, or **MLEs**. The probability of each observation is given as:

$$\mathbf{Y}_i \approx N\left(\mathbf{X}\boldsymbol{\beta}, \sigma^2\right) \qquad f\left(x_i, \boldsymbol{\theta}\right) = P\left(y_i\right) = \frac{1}{\sqrt{2\pi\sigma}} \exp\left(-\frac{\left(y_i - \boldsymbol{X\beta}\right)^2}{2\sigma^2}\right)$$

- Likelihood function is the joint density of observing the sample data from a statistical distribution with parameter vector

$$f\left(x_1, x_2, \ldots, x_n, \boldsymbol{\theta}\right) = \prod_{i=1}^{n} f\left(x_i, \boldsymbol{\theta}\right) = L(\boldsymbol{\theta} \mid \mathbf{X})$$

Maximum Likelihood Estimation

- For the regression model, the likelihood function for a sample of n independent, identically, and normally distributed disturbances is given by:

$$L = \left(2\pi\sigma^2\right)^{-\frac{n}{2}} EXP\left[-\frac{1}{2\sigma^2}\sum_{i=1}^{n}\left(Y_i - X_i^T\beta\right)^2\right] = \left(2\pi\sigma^2\right)^{-\frac{n}{2}} EXP\left[-\frac{1}{2\sigma^2}(Y - X\beta)^T(Y - X\beta)\right]$$

- As is usually the case, the logarithm of Equation, or the log likelihood, is simpler to solve than the likelihood function itself, so taking the log of L yields:

$$LN(L) = LL = -\frac{n}{2}LN(2\pi) - \frac{n}{2}LN\left(\sigma^2\right) - \frac{1}{2\sigma^2}(\mathbf{Y} - \mathbf{X\beta})^T(\mathbf{Y} - \mathbf{X\beta})$$

Inference in Regression Analysis

- Consider again the study of AADT in Minnesota. Interest centers on the development of confidence intervals and hypothesis tests on some of the parameters estimated in Example 3.1

Least Squares Estimated Parameters (Example 3.2)

Parameter	Parameter Estimate	Standard Error of Estimate	t-value	$P(>\|t\|)$
Intercept	–26234.490	4935.656	–5.314	<0.0001
CNTYPOP	0.029	0.005	5.994	<0.0001
NUMLANES	9953.676	1375.433	7.231	<0.0001
FUNCLASS1	885.384	5829.987	0.152	0.879
FUNCLASS2	4127.560	3345.418	1.233	0.220
FUNCLASS3	35453.679	4530.652	7.825	<0.0001

Manipulating Variables in Regression

- **Standardized** regression models allow for direct comparison of the relative importance of independent variables

- Often interest is focused on the **relative impacts** of independent variables on the response variable Y

- Two independent variables in a model describing the *expected number of daily trip-chains* were *number of children* and *household income*

- Households may have children ranging from 0 to 8, while income may range from $5000 to $500,000, making it difficult to make a useful comparison between the relative impact of these variables

Manipulating Variables in Regression

- The standardized regression model is obtained by **standardizing** all **independent variables**.

$$X_1' = \frac{X_1 - \overline{X}}{s\{X_1\}}$$

- The standardized variables are created with expected values equal to 0 and variances equal to 1.

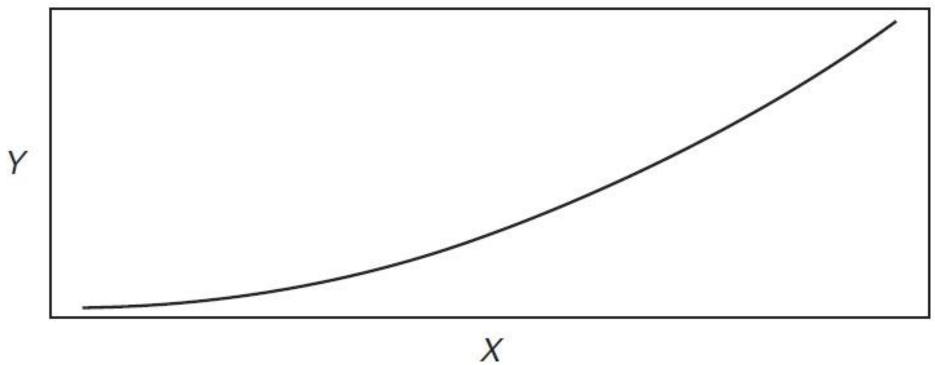
Manipulating Variables in Regression

- **Nonlinear relationships** can be accommodated within the linear regression framework by **Transformations**

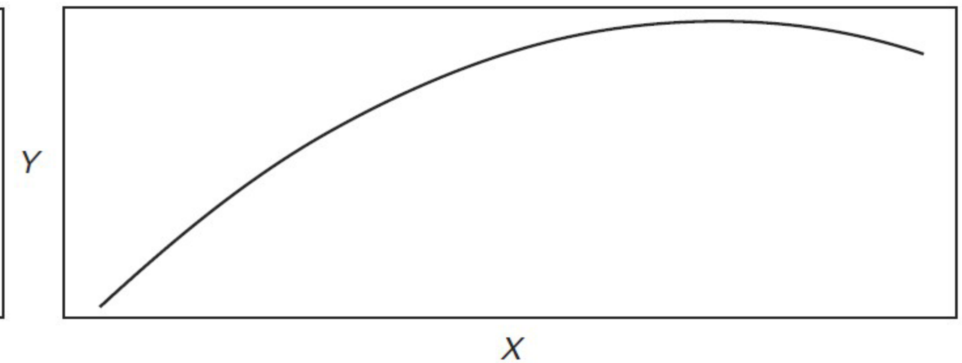
$$\hat{Y} = \frac{X}{B_0X + B_1 + eX} = \frac{X}{\alpha X + \lambda + Xe} \quad \Rightarrow \quad E\left[\frac{1}{\hat{Y}}\right] = B_0 + B_1\frac{1}{X} + e$$

$$\hat{Y} = EXP^{B_0}EXP^{\frac{B_1}{X}}EXP^{e} \quad \Rightarrow \quad LN\left(\hat{Y}\right) = B_0 + \frac{B_1}{X} + e$$

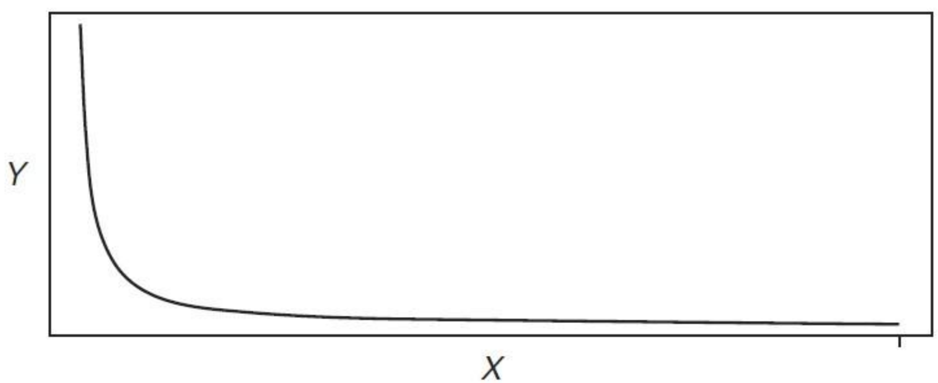
Manipulating Variables in Regression



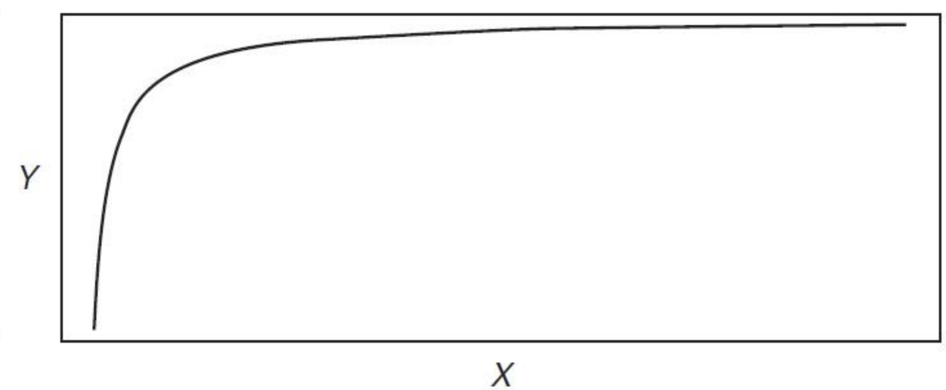
$Y = \alpha + \beta X + \gamma X^2$, where $\alpha > 0$, $\beta > 0$, $\gamma > 0$



$Y = \alpha + \beta X + \gamma X^2$, where $\alpha > 0$, $\beta > 0$, $\gamma < 0$

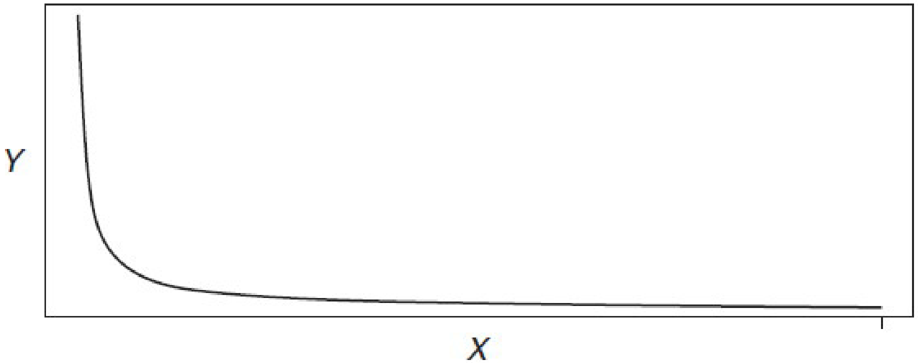


$Y = X/(\alpha + \beta X)$, where $\alpha > 0$

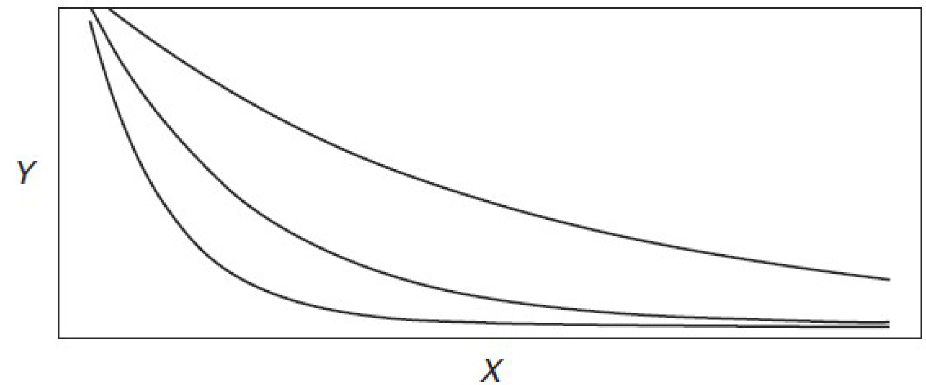


$Y = X/(\alpha + \beta X)$, where $\alpha < 0$

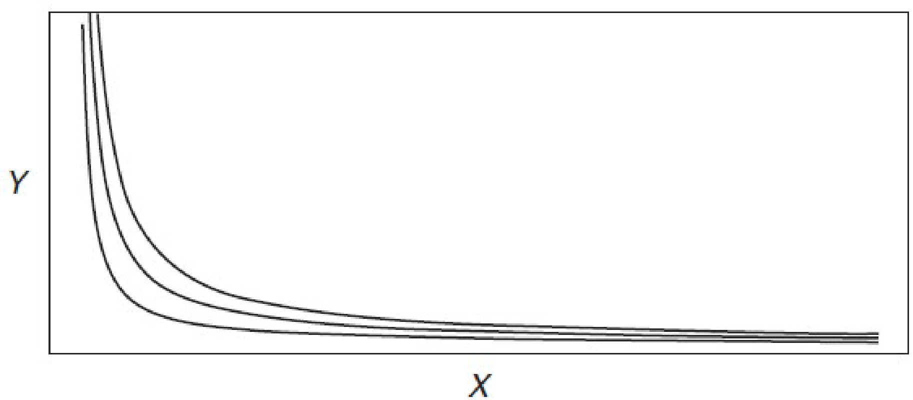
Manipulating Variables in Regression



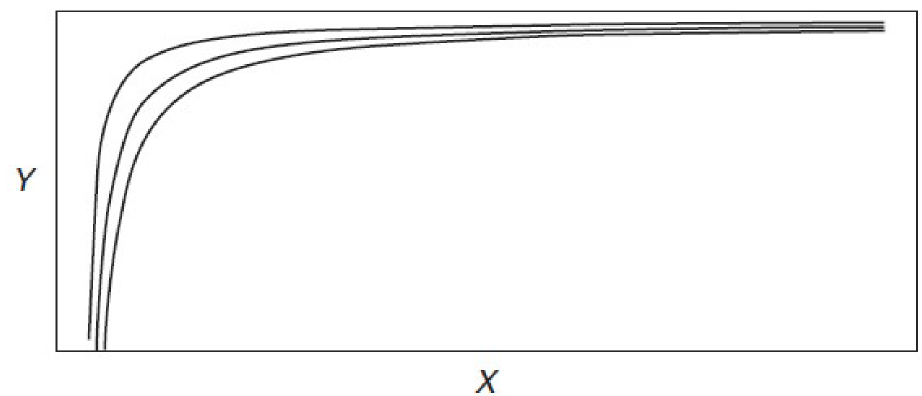
$Y = \alpha EXP(\beta X)$, where $\beta > 0$



$Y = \alpha EXP(\beta X)$, where $\beta < 0$

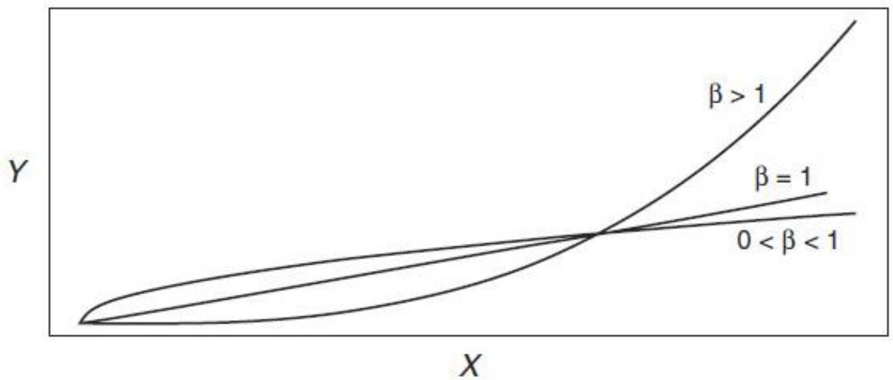


$Y = \alpha EXP(\beta / X)$, where $\beta > 0$

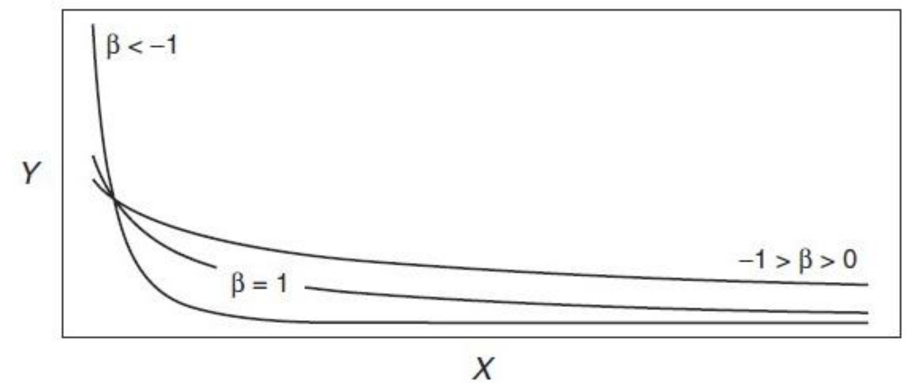


$Y = \alpha EXP(\beta / X)$, where $\beta < 0$

Manipulating Variables in Regression



$Y = \alpha X\beta$, where $\beta > 0$



$Y = \alpha X\beta$, where $\beta < 0$

Manipulating Variables in Regression

- Nominal scale variables need transformed into indicator Variables
- Qualitative variables such as *roadway functional class*, *gender*, *attitude toward transit*, and *trip purpose*
- For **nominal scale variables**, **m-1 indicator variables** must be created to represent all **m levels** of the variable in the regression model.

Manipulating Variables in Regression

- Estimate a Single Beta Parameter: The assumption in this approach is that the marginal effect is equivalent across increasing levels of the variable
- Estimate Beta Parameter for Ranges of the Variable: Consider the variable NUMLANES used in previous chapter examples. Although the intervals between levels of this variable are equivalent, it may be believed that a fundamentally different effect on AADT exists for different levels of NUMLANES.

$$Ind_1 = \begin{cases} NUMLANES \text{ if } 1 \le NUMLANES \le 2 \\ 0 \text{ otherwise} \end{cases} \qquad Ind_2 = \begin{cases} NUMLANES \text{ if } NUMLANES > 2 \\ 0 \text{ otherwise} \end{cases}$$

- These two indicator variables would allow the estimation of two parameters, one for the lower range of the variable NUMLANES and one for the upper range

Manipulating Variables in Regression

- Estimate a Single Beta Parameter for **m-1** of the **m Levels** of the Variable: The justification for this approach is that each level of the variable has a unique marginal effect on the response

$$Ind_1 = \begin{cases} 1 \text{ if } NUMLANES = 1 \\ 0 \text{ otherwise} \end{cases}$$

$$Ind_2 = \begin{cases} 1 \text{ if } NUMLANES = 2 \\ 0 \text{ otherwise} \end{cases}$$

$$Ind_3 = \begin{cases} 1 \text{ if } NUMLANES = 3 \\ 0 \text{ otherwise} \end{cases}$$

Manipulating Variables in Regression

- **Interactions** in regression models represent a **combined** or **synergistic** effect of two or more variables. That is, the response variable depends on the joint values of two or more variables.

$$\hat{Y} = B_0 + B_1X_1 + B_2X_2 + B_3X_3 + B_4X_4 .$$

$$\text{level 1:} \quad X_2 = X_3 = X_4 = 0 \qquad \hat{Y} = B_0 + B_1X_1$$

$$\text{level 2:} \quad X_2 = 1 \qquad \hat{Y} = B_0 + B_1X_1 + B_2X_2 = (B_0 + B_2) + B_1X_1$$

$$\text{level 3:} \quad X_3 = 1 \qquad \hat{Y} = B_0 + B_1X_1 + B_3X_3 = (B_0 + B_3) + B_1X_1$$

$$\text{level 4:} \quad X_4 = 1 \qquad \hat{Y} = B_0 + B_1X_1 + B_4X_4 = (B_0 + B_4) + B_1X_1.$$

Manipulating Variables in Regression

$$\hat{Y} = B_0 + B_1X_1 + B_2X_2 + B_3X_3 + B_4X_4 .$$

$$\text{level 1:} \quad X_2 = X_3 = X_4 = 0 \qquad \hat{Y} = B_0 + B_1X_1$$

$$\text{level 2:} \quad X_2 = 1 \qquad \hat{Y} = B_0 + B_1X_1 + B_2X_2 = (B_0 + B_2) + B_1X_1$$

$$\text{level 3:} \quad X_3 = 1 \qquad \hat{Y} = B_0 + B_1X_1 + B_3X_3 = (B_0 + B_3) + B_1X_1$$

$$\text{level 4:} \quad X_4 = 1 \qquad \hat{Y} = B_0 + B_1X_1 + B_4X_4 = (B_0 + B_4) + B_1X_1 .$$

- Depending on which of the indicator variables is coded as 1, the slope of the regression line with respect to **X1 remains fixed**, while the **Y-intercept** parameter changes by the amount of the parameter of the indicator variable.

Manipulating Variables in Regression

- Suppose that each indicator variable is thought to interact with the variable X1. That is, each level of the indicator variable has a unique effect on Y when interacted with X1.

$$\hat{Y} = B_0 + B_1X_1 + B_2X_2 + B_3X_3 + B_4X_4 + B_5X_2X_1 + B_6X_3X_1 + B_7X_4X_1$$

level 1: $X_2 = X_3 = X_4 = 0$ $\quad \hat{Y} = B_0 + B_1X_1$

level 2: $X_2 = 1$ $\quad \hat{Y} = B_0 + B_1X_1 + B_2X_2 + B_5X_2X_1 = (B_0 + B_2) + (B_1 + B_5)X_1$

level 3: $X_3 = 1$ $\quad \hat{Y} = B_0 + B_1X_1 + B_3X_3 + B_6X_3X_1 = (B_0 + B_3) + (B_1 + B_6)X_1$

level 4: $X_4 = 1$ $\quad \hat{Y} = B_0 + B_1X_1 + B_4X_4 + B_7X_4X_1 = (B_0 + B_4) + (B_1 + B_7)X_1$

Manipulating Variables in Regression

$$\hat{Y} = B_0 + B_1X_1 + B_2X_2 + B_3X_3 + B_4X_4 + B_5X_2X_1 + B_6X_3X_1 + B_7X_4X_1$$

level 1: $X_2 = X_3 = X_4 = 0$ $\quad \hat{Y} = B_0 + B_1X_1$

level 2: $X_2 = 1$ $\quad \hat{Y} = B_0 + B_1X_1 + B_2X_2 + B_5X_2X_1 = (B_0 + B_2) + (B_1 + B_5)X_1$

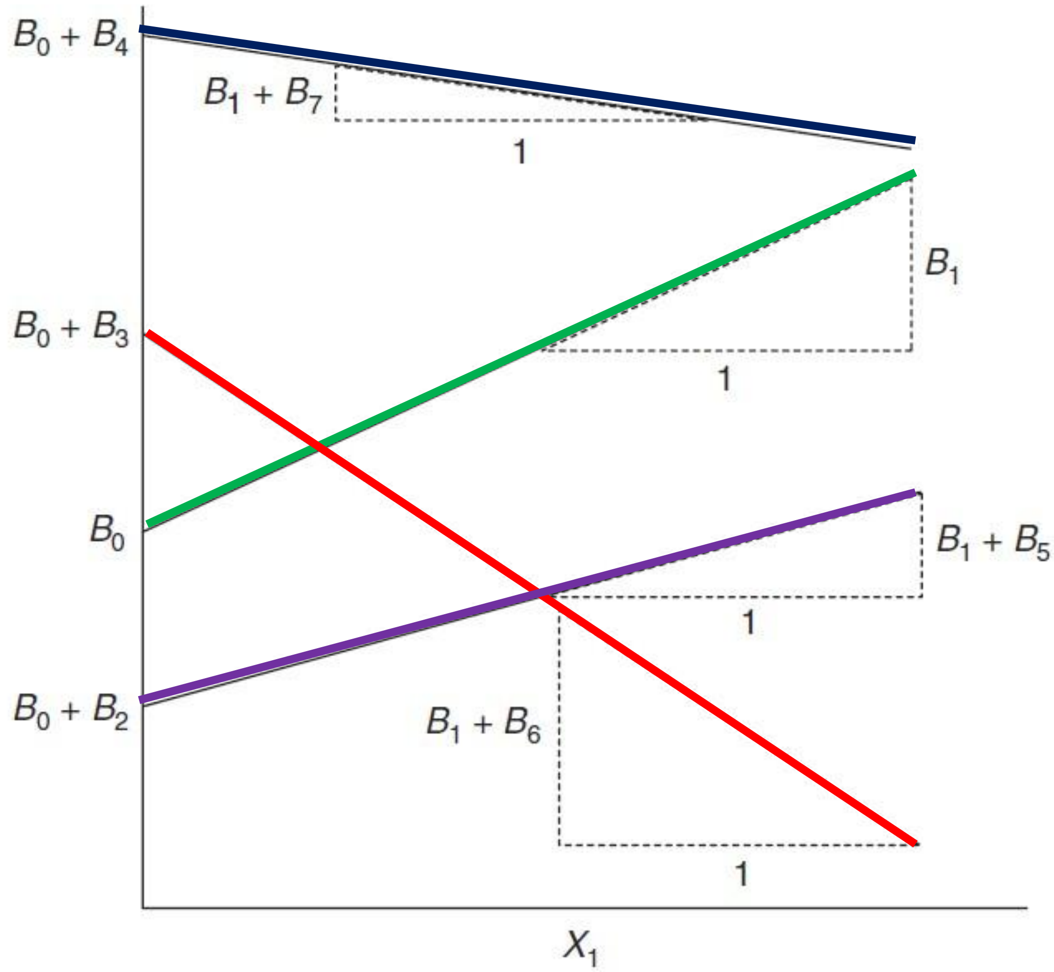
level 3: $X_3 = 1$ $\quad \hat{Y} = B_0 + B_1X_1 + B_3X_3 + B_6X_3X_1 = (B_0 + B_3) + (B_1 + B_6)X_1$

level 4: $X_4 = 1$ $\quad \hat{Y} = B_0 + B_1X_1 + B_4X_4 + B_7X_4X_1 = (B_0 + B_4) + (B_1 + B_7)X_1$

- Each level of the indicator variable now has an effect on both the **Y-intercept** and **slope** of the regression function with respect to the variable X1.

Manipulating Variables in Regression

$$\hat{Y} = B_0 + B_1X_1 + B_2X_2 + B_3X_3 + B_4X_4 + B_5X_2X_1 + B_6X_3X_1 + B_7X_4X_1$$



Manipulating Variables in Regression

- When interactions are between two or more continuous variables the regression equation becomes more complicated.

$$\hat{Y} = B_0 + B_1X_1 + B_2X_2 + B_3X_1X_2$$

- The regression function indicates that the **relationship between X_1 and Y** is dependent on the value of **X_2**, and conversely, that the **relation between X_2 and Y** is dependent on the value of **X_1**.

- There are **numerous cases** when the effect of one variable on Y depends on the value of one or more independent variables.

- Although they **may not explain** a great deal of the **variability** in the response, they may represent **important theoretical** aspects of the relation and should be **included** in the regression.

Model Goodness-of-Fit Measures

- GOF: **R-squared**, and **adjusted R-squared**.

- To develop the R-squared GOF statistic, some basic notions are required:

- The sum of square errors (disturbances) is given by:

$$\mathrm{SSE} = \sum_{i=1}^{n} \left(Y_i - \hat{Y}_i \right)^2$$

- the regression sum of squares is given by

$$\mathrm{SSR} = \sum_{i=1}^{n} \left(\hat{Y}_i - \overline{Y} \right)^2$$

Model Goodness-of-Fit Measures

- The total sum of squares is given by

$$\text{SST} = \sum_{i=1}^{n} \left(Y_i - \overline{Y}\right)^2$$

- It also can be shown algebraically that **SST = SSR + SSE.**
- The coefficient of determination, R-squared, is defined as

$$R^2 = \frac{[\text{SST} - \text{SSE}]}{\text{SST}} = \frac{\text{SSR}}{\text{SST}} = 1 - \frac{\text{SSE}}{\text{SST}}.$$

- The coefficient of determination, R-squared, is defined as

$$R^2{}_{\text{adjusted}} = 1 - \frac{\dfrac{\text{SSE}}{n-p}}{\dfrac{\text{SST}}{n-1}} = 1 - \left(\frac{n-1}{n-p}\right)\frac{\text{SSE}}{\text{SST}}$$

Model Goodness-of-Fit Measures

- The R^2 and $R^2_{adjusted}$ measures provide only **relevant comparisons** with previous models that have been estimated on the phenomenon under investigation
- The absolute values of R^2 and R^2 adjusted measures **are not sufficient** measures to judge the quality of a model.
- Relatively large values of R^2 and R^2 adjusted can be caused by data artifacts.
- The R^2 value is 0.8947. The collection of independent variables accounts for about 89% of the uncertainty or variation in AADT.
- This is considered to be a good result because previous studies have achieved R-squared values of around 70%. It is only with respect to other models on the same phenomenon that R-squared comparisons are meaningful.